MPG No. 2.3.5 $\bar{1}'$ [Type III, triclinic] [T tensor]

* Rank 0 tensor. * Rank 1 tensor.

$$\begin{bmatrix} T_x & T_y & T_z \end{bmatrix}$$

$$T_x = \mathbb{T}_x^{(1)}$$
$$T_y = \mathbb{T}_y^{(1)}$$
$$T_z = \mathbb{T}_z^{(1)}$$

* Rank 2 tensor (s). * Rank 2 tensor (a). * Rank 3 tensor (ss).

$$\begin{bmatrix} T_{xxx} & T_{xxy} & T_{xxz} \\ T_{yyx} & T_{yyy} & T_{yyz} \\ T_{zzx} & T_{zzy} & T_{zzz} \\ T_{yzx} & T_{yyz} & T_{zzy} \\ T_{xxz} & T_{yzx} & T_{zzx} \\ T_{xxy} & T_{yyx} & T_{yzx} \end{bmatrix}$$

$$T_{xxx} = 3\mathbb{T}_x^{(1)} + 2\mathbb{T}_x^{\alpha(1)}$$
$$T_{xxy} = \mathbb{T}_y^{(1)} - \mathbb{T}_y^{\alpha(1)} - \mathbb{T}_y^{\beta(1)}$$
$$T_{xxz} = \mathbb{T}_z^{(1)} - \mathbb{T}_z^{\alpha(1)} + \mathbb{T}_z^{\beta(1)}$$
$$T_{yyx} = \mathbb{T}_x^{(1)} - \mathbb{T}_x^{\alpha(1)} + \mathbb{T}_x^{\beta(1)}$$
$$T_{yyy} = 3\mathbb{T}_y^{(1)} + 2\mathbb{T}_y^{\alpha(1)}$$
$$T_{yyz} = \mathbb{T}_z^{(1)} - \mathbb{T}_z^{\alpha(1)} - \mathbb{T}_z^{\beta(1)}$$
$$T_{zzx} = \mathbb{T}_x^{(1)} - \mathbb{T}_x^{\alpha(1)} - \mathbb{T}_x^{\beta(1)}$$
$$T_{zzy} = \mathbb{T}_y^{(1)} - \mathbb{T}_y^{\alpha(1)} + \mathbb{T}_y^{\beta(1)}$$
$$T_{zzz} = 3\mathbb{T}_z^{(1)} + 2\mathbb{T}_z^{\alpha(1)}$$
$$T_{yzx} = \mathbb{T}_{xyz}^{(1)}$$

* Rank 3 tensor (sa).

$$\begin{bmatrix} 0 & T_{xxy} & T_{xxz} \\ T_{yyx} & 0 & T_{yyz} \\ T_{zzx} & T_{zzy} & 0 \\ T_{yzx} & -\frac{T_{yyz}}{2} & -\frac{T_{zzy}}{2} \\ -\frac{T_{xxz}}{2} & T_{zxy} & -\frac{T_{zzx}}{2} \\ -\frac{T_{xxy}}{2} & -\frac{T_{yyx}}{2} & -T_{yzx} - T_{zxy} \end{bmatrix}$$

$$T_{xxy} = -2\mathbb{M}^{(1)}_{xz} + 2\mathbb{T}^{(2)}_{y}$$
$$T_{xxz} = 2\mathbb{M}^{(1)}_{xy} + 2\mathbb{T}^{(2)}_{z}$$
$$T_{yyx} = 2\mathbb{M}^{(1)}_{yz} + 2\mathbb{T}^{(2)}_{x}$$
$$T_{yyz} = -2\mathbb{M}^{(1)}_{xy} + 2\mathbb{T}^{(2)}_{z}$$
$$T_{zzx} = -2\mathbb{M}^{(1)}_{yz} + 2\mathbb{T}^{(2)}_{x}$$
$$T_{zzy} = 2\mathbb{M}^{(1)}_{xz} + 2\mathbb{T}^{(2)}_{y}$$
$$T_{yzx} = 3\mathbb{M}^{(1)}_{u} + \mathbb{M}^{(1)}_{v}$$
$$\frac{T_{yyz}}{2} = -\mathbb{M}^{(1)}_{xy} + \mathbb{T}^{(2)}_{z}$$
$$\frac{T_{zzy}}{2} = \mathbb{M}^{(1)}_{xz} + \mathbb{T}^{(2)}_{y}$$
$$\frac{T_{xxz}}{2} = \mathbb{M}^{(1)}_{xy} + \mathbb{T}^{(2)}_{z}$$
$$T_{zxy} = -3\mathbb{M}^{(1)}_{u} + \mathbb{M}^{(1)}_{v}$$
$$\frac{T_{zzx}}{2} = -\mathbb{M}^{(1)}_{yz} + \mathbb{T}^{(2)}_{x}$$
$$\frac{T_{xxy}}{2} = -\mathbb{M}^{(1)}_{xz} + \mathbb{T}^{(2)}_{y}$$
$$\frac{T_{yyx}}{2} = \mathbb{M}^{(1)}_{yz} + \mathbb{T}^{(2)}_{x}$$

* Rank 3 tensor (a).

$$\begin{bmatrix} T_{yzx} & T_{yzy} & T_{yzz} \\ T_{zxx} & T_{zxy} & T_{zxz} \\ T_{xyx} & T_{xyy} & T_{xyz} \end{bmatrix}$$

$$T_{yzx} = \mathbb{M}^{(1)}_{0} - \mathbb{M}^{(2)}_{u} + \mathbb{M}^{(2)}_{v}$$
$$T_{yzy} = \mathbb{M}^{(2)}_{xy} + \mathbb{T}^{(3)}_{z}$$
$$T_{yzz} = \mathbb{M}^{(2)}_{xz} - \mathbb{T}^{(3)}_{y}$$
$$T_{zxx} = \mathbb{M}^{(2)}_{xy} - \mathbb{T}^{(3)}_{z}$$
$$T_{zxy} = \mathbb{M}^{(1)}_{0} - \mathbb{M}^{(2)}_{u} - \mathbb{M}^{(2)}_{v}$$
$$T_{zxz} = \mathbb{M}^{(2)}_{yz} + \mathbb{T}^{(3)}_{x}$$
$$T_{xyx} = \mathbb{M}^{(2)}_{xz} + \mathbb{T}^{(3)}_{y}$$
$$T_{xyy} = \mathbb{M}^{(2)}_{yz} - \mathbb{T}^{(3)}_{x}$$
$$T_{xyz} = \mathbb{M}^{(1)}_{0} + 2\mathbb{M}^{(2)}_{u}$$

* Rank 4 tensor (sss). * Rank 4 tensor (ssa). * Rank 4 tensor (aas). * Rank 4 tensor (aaa). * Rank 4 tensor (sa). * Rank 4 tensor (as). * Rank 4 tensor (s). * Rank 4 tensor (a). * Rank 4 tensor (t).